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19251177100

QUESTIONS 3 / 5 completed

Merge Sort

MERGE SORT

Apply Merge Sort to the array 38, 27, 43, 3, 9, 82, 10.

PROGRAM EDITOR C ✓ Saved Run Save

```
1 #include <stdio.h>
2
3 void merge(int arr[], int left, int mid, int right) {
4     int n1 = mid - left + 1;
5     int n2 = right - mid;
6
7     int L[n1], R[n2];
8
9     for (int i = 0; i < n1; i++)
10         L[i] = arr[left + i];
11     for (int j = 0; j < n2; j++)
12         R[j] = arr[mid + 1 + j];
13
14     int i = 0, j = 0, k = left;
15
16     while (i < n1 && j < n2) {
17         if (L[i] <= R[j]) {
18             arr[k] = L[i];
19             i++;
20         } else {
21             arr[k] = R[j];
```

OUTPUT

3 9 10 27 38 43 82

INPUT

38 27 43 3 9 82 10

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```
21         arr[k] = R[j];
22         j++;
23     }
24     k++;
25 }
26
27 while (i < n1) {
28     arr[k] = L[i];
29     i++;
30     k++;
31 }
32
33 while (j < n2) {
34     arr[k] = R[j];
35     j++;
36     k++;
37 }
38 }
39
40 void mergeSort(int arr[], int left, int right) {
41     if (left < right) {
```

OUTPUT

3 9 10 27 38 43 82

INPUT

38 27 43 3 9 82 10

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QUESTIONS 3 / 5 completed

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Apply Merge Sort to the array 38, 27, 43, 3, 9, 82, 10.

PROGRAM EDITOR

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```
41 if (left < right) {
42     int mid = left + (right - left) / 2;
43
44     mergeSort(arr, left, mid);
45     mergeSort(arr, mid + 1, right);
46
47     merge(arr, left, mid, right);
48 }
49 }
50
51 int main() {
52     int arr[] = {38, 27, 43, 3, 9, 82, 10};
53     int n = sizeof(arr) / sizeof(arr[0]);
54
55     mergeSort(arr, 0, n - 1);
56
57     for (int i = 0; i < n; i++)
58         printf("%d ", arr[i]);
59
60     return 0;
61 }
```

OUTPUT

3 9 10 27 38 43 82

INPUT

38 27 43 3 9 82 10